

ESA MARIA SHYJU

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Location: Thiruvananthapuram, Kerala, India

PROFESSIONAL SUMMARY

Motivated and detail-oriented final-year Computer Science student with a strong foundation in Machine Learning, Artificial Intelligence, and Full-Stack Development. Skilled in building scalable web applications and AI-powered solutions using modern technologies. Seeking opportunities to apply my technical knowledge, problem-solving abilities, and passion for innovation to contribute effectively to organizational growth while continuously enhancing my professional skills.

KEY EXPERTISE

Full Stack Web Development , Mern Stack Development, Artificial Intelligence, Machine Learning, LLM fine tuning, Python Programming, Java Programming, Problem Solving & Analytical Thinking, DBMS, Data Structures

EDUCATION

Mar Baselios College of Engineering and Technology (Autonomous)	2023-2027
B.Tech. - Computer Science & Engineering CGPA: 9.32 / 10	
St. Mary's Hss,Pattom, Thiruvananthapuram	2023
12th KBHSE Percentage: 96 / 100	
Infant Jesus Central School, Thiruvananthapuram	2021
10th CBSE Percentage: 94 / 100	

INTERNSHIPS

ICT Academy of Kerala	15 Dec, 2024 - 31 Jan, 2025
Full Stack Development Intern Designed and developed MERN stack web applications with React, Node.js, Express, and MongoDB, implementing RESTful APIs, authentication, and responsive user interfaces.	
Intel Unnati Program	20 May, 2024 - 20 Jul, 2024
AI & Solution Design Intern Fine-tuned Large Language Models using Python and Hugging Face to develop a custom AI chatbot, gaining experience in NLP, prompt engineering, model evaluation, deployment, and CPU inference optimization.	

PROJECTS

FPGA-Based Hardware Acceleration System	20 May, 2024 - 20 Jul, 2024
Developed an FPGA-based hardware acceleration system focused on improving computational throughput through optimized architectural execution. Designed and implemented sequential and parallel processing approaches to evaluate execution efficiency and latency reduction. Integrated timing control mechanisms, hardware mapping strategies, and real-time visualization to demonstrate measurable performance improvements through hardware-level optimization.	

Library Management System

15 Dec, 2024 - 31 Jan, 2025

Worked as a Full Stack Development Intern focusing on designing and developing scalable web applications using the MERN stack. Built responsive frontend interfaces and implemented backend services with RESTful APIs, authentication, and database integration. Contributed to application optimization, collaborative development workflows, and end-to-end feature implementation while following modern software engineering practices.

Custom AI Chatbot using LLM Fine-Tuning

01 May, 2024 - 31 Jul, 2024

Worked as a Full Stack Development Intern focusing on designing and developing scalable web applications using the MERN stack. Built responsive frontend interfaces and implemented backend services with RESTful APIs, authentication, and database integration. Contributed to application optimization, collaborative development workflows, and end-to-end feature implementation while following modern software engineering practices.

Network Intrusion Detection using ANN

10 Feb, 2026 - 20 Apr, 2026

Built an intelligent network intrusion detection system using Artificial Neural Networks to classify and detect malicious network activity. Developed complete machine learning pipelines including preprocessing, feature engineering, model training, and evaluation using benchmark datasets. Designed an analytics dashboard to visualize detection insights and improve monitoring accuracy for security focused decision making.

ASSESSMENTS/CERTIFICATIONS

Deep Learning - IIT Ropar

Provider: NPTEL

Key Skills: Deep Learning , ANN, AI

Completed a 12-week certification program offered by IIT Ropar through NPTEL, funded by the Ministry of Education, Govt. of India. Gained hands-on knowledge in deep learning, neural networks, and AI applications, with exposure to both theoretical foundations and practical problem-solving. Certified among thousands of candidates nationwide.

Introduction to Large Language Models (LLMs)

Provider: NPTEL

Key Skills: LLM , Transformer architectures, Fine-tuning methods, Natural language processing

Completed a 12-week certification program on large language models, covering theoretical foundations and practical applications in natural language processing. Gained exposure to transformer architectures, fine-tuning techniques, and real-world AI use cases. Certified through NPTEL, funded by the Ministry of Education, Govt. of India.

Artificial Intelligence Primer Certification

Provider: Infosys Springboard

Key Skills: AI, Machine Learning, ANN, Deep Learning

Completed a foundational certification program on artificial intelligence, covering core concepts, applications, and industry practices. Developed understanding of machine learning basics, AI ethics, and real-world use cases. Certified by Infosys Springboard, demonstrating commitment to continuous learning in emerging technologies.

PERSONAL DETAILS

Languages: English, Malayalam

Date of Birth: 02 JUNE, 2005

Marital Status: Single

GitHub: github.com/esamaria

LinkedIn: <https://www.linkedin.com/in/esa-maria-shyju>